How-To Library

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How To Use the Custom Metadata Validation Utility with Enterprise Data Catalog

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Abstract

This article provides information about tuning Enterprise Data Catalog performance. Tuning Enterprise Data Catalog performance involves tuning parameters for metadata ingestion, ingestion database, search, and tuning data profiling. The performance of Enterprise Data Catalog depends on the size of data that needs to be processed.

Supported Versions

- Enterprise Data Catalog 10.4.1

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Overview

In Enterprise Data Catalog, custom metadata signifies metadata ingested to the catalog from data sources for which Enterprise Data Catalog does not provide metadata extraction support by default. Enterprise Data Catalog, by default, provides models for multiple data sources from which you can extract metadata. To ingest custom metadata in the catalog, you must define a custom model as the first step. You can define a custom model in Catalog Administrator. The model defines the structure of the metadata stored in the catalog. After you define the custom model, you must create a custom resource type in Catalog Administrator based on the custom model. Custom metadata integration signifies the process to ingest custom metadata in the catalog.

You can use the Java-based Validation Command Line Utility to validate the custom metadata before you ingest the custom metadata in the catalog.

Custom Metadata Integration Process

Custom metadata integration in Enterprise Data Catalog involves the following steps:

1. Create the custom model.
2. Create the custom resource type.
3. Create the custom resource.
4. Add the metadata from the custom source in the following files:
   - objects.csv. Use this file to enter data based on the class details in the data source.
   - links.csv. Use this file to enter association details of objects in the data source.
5. Provide the ZIP file that includes the objects.csv and links.csv files when you configure the custom resource.

Note: See the Informatica Enterprise Data Catalog Administrator Guide for more information about Custom Metadata Integration.

Custom Metadata Validation Utility

Before you ingest the custom metadata in the catalog, you can use the custom metadata validation utility to validate the syntax and semantics of the custom metadata included in the objects.csv and links.csv files. The utility saves time by identifying validation errors before you ingest metadata into the catalog from unsupported sources.

Note: The custom metadata validation utility does not validate the metadata if `objects.csv`, `links.csv` and `lineage.csv` files include context metadata.

You can run the standalone utility on a machine running on Microsoft Windows or Linux operating system.

The utility generates an HTML or CSV validation report that you can use to verify the validity of the custom metadata. The report provides a list of validation errors if there are any, with error codes and lists the errors with hints that you can use to resolve the errors.

Downloading the Utility

To download the custom-scanner-validator.zip file that includes the Validation Command Line Utility, contact Informatica Global Customer Support.

Prerequisites to Run the Utility

Before you run the utility, verify the following prerequisites :

- Install Java Runtime Environment (JRE) 7 or later on the machine where you want to run the utility.
- Ensure that the JAVA_HOME environment variable points to the JRE installed on the machine. You need not change the `JAVA_HOME` environment variable of the machine.
- Create a ZIP file that includes the `objects.csv` and one `links.csv` file.
- Specify if you want to generate the validation report in CSV or HTML format. Specify `html` or `csv` in the `report.format=` section of the `app.properties` file.

Validations Performed by the Utility

The utility validates the following attributes of the custom metadata and model:

- Syntax. Includes validating the syntax of the headers, separators, object IDs, and valid characters. For example, the object IDs must only contain alphanumeric characters and the special characters supported are forward slash (/) and underscore (_).
- Class. Includes validating the class type of the object with the custom model. For example, the object class mentioned in the `objects.csv` file must be present in the custom model.
- Association. Includes validating the associations that link the objects as shown in the following list:
  - The `fromObjectID` and `toObjectID` mentioned in `links.csv` must be present in the `objects.csv` file.
  - The association class mentioned in the `links.csv` file must be registered with the core model.
  - The class type that includes `fromObjectID` and `toObjectID` must match the class type that includes `fromClass` and `toClass` in the association.
  - One `core.ResourceParentChild` association entry must be present in the `links.csv` file. `fromObjectID` can be empty for the association, but the `toObjectID` must be the root object.

- Data consistency. Includes validating the objects based on the attribute type, verifying the presence of orphan objects in the metadata, and validating the presence of objects referenced in the `links.csv` file with the `objects.csv` file as shown in the following list:
  - The number of fields in each record must match the number of columns in the header.
  - The mandatory attributes for a model must be present with valid values for each object in the `objects.csv` file.
  - The data type of the attributes must have a valid literal that is compliant to the data type.
  - The data type of the object attribute must be present in the core model.
  - The value specified for an object attribute must be a literal of the data type mentioned in the core model.

Running the Utility

You can run the utility on a Microsoft Windows machine or Linux machine. After you download the custom-scanner-validator.zip file, extract the contents of the ZIP file to your machine.

The following files and directories are extracted under the custom-scanner-validator directory on your machine:

- validate.bat
- validate.sh
- /conf/app.properties
- /lib/custom-scanner-validator.jar

Linux

1. Go to the `custom-scanner-validator` directory on your machine.
2. Run the `validate.sh` script as shown in the following sample: `./validate.sh --model <path to the ZIP file that includes the custom model> --metadata <path to the ZIP file that includes the objects.csv and the links.csv files>`

The utility generates an HTML or CSV validation report in the in the `custom-scanner-validator/output` directory.

Windows

1. Go to the `custom-scanner-validator` folder on your machine.
2. Run the `validate.bat` batch file as shown in the following sample: `validate.bat --model <path to the ZIP file that includes the custom model> --metadata <path to the ZIP file that includes the objects.csv and the links.csv files>`

The utility generates an HTML or CSV validation report in the in the `custom-scanner-validator/output` folder.

Utility Parameters

The following table describes the parameters that you can use with the utility:

Parameter	Mandatory or Optional	Description
--model	Mandatory	Path to the ZIP file that includes the custom model and core model XML files.
--metadata	Mandatory	Path to the ZIP file that includes the `objects.csv` and the `links.csv` files.

Parameter	Mandatory or Optional	Description
--stop-on-error	Optional	Set the parameter to `Yes` if you want the utility to stop running after it encounters a validation error. Default is `No`.
--mem	Optional	Specify the memory required to run the script based on the amount of metadata that you plan to ingest into the catalog. You can choose any of the following options: - low. Signifies that the script can use a minimum of 512 MB of memory and a maximum of 2 GB of memory. - medium. Signifies that the script can use a minimum of 2 GB of memory and a maximum of 4 GB of memory. - high. Signifies that the script can use a minimum of 2 GB of memory and a maximum of 8 GB of memory.

Note: You can use the `--help` option with the utility to view the description of the parameters.

Sample Command Usage

The following sample command shows the usage of all the parameters with the Validation Command Line Utility on a Linux machine:

```
./validate.sh --model /home/inputset/model.zip --metadata /home/inputset/metadata.zip --stop-on-error No --mem low
```

Validation Report

The following image shows the validation report generated in HTML format:

Enterprise Data Catalog Custom Scanner Validation Report

Scanner Name : custom, **Report Generated On :** Mon, 21 Jan 2019 13:08:18 IST

Summary Report (Total Errors: 14)

Error Code	Error Description	File	Error Count	Resolution Hint
ERR_OBJECTS_001	Invalid Object Class	objects.csv	2	Register the referred class in the model or change the class type for the object referred.
ERR_OBJECTS_002	Invalid ObjectID Format.	objects.csv	1	Object ID should be of format 'DataSource/DataSet/[DataSet].../DataElement' and contain only alphanumeric characters, forward slash and underscore.
ERR_OBJECTS_003	Invalid Parent Object.	objects.csv	1	Make an entry for the parent object in objects.csv or remove the offending entry.

Detailed Report

Error Code	File	Line No	Error Description	Resolution Hint
ERR_OBJECTS_006	objects.csv	4	Actual number of fields (46) do not match expected number of fields in header (48)	Check for missing values, extra or missing separator (comma)
ERR_OBJECTS_002	objects.csv	5	ObjectId steam-thinker-226107#$@@#@@ has invalid characters	Object ID should be of format 'DataSource/DataSet/[DataSet].../DataElement' and contain only alphanumeric characters, forward slash and underscore.
ERR_OBJECTS_005	objects.csv	6	Invalid value lol for attribute google.customBigquery.autodetect having type Boolean	Correct the attribute value to match the data type and other constraints.
ERR_OBJECTS_005	objects.csv	8	Invalid value lol for attribute google.customBigquery.numOfRows having type Integer	Correct the attribute value to match the data type and other constraints.

The Summary Report lists the error code, the error description, the file in which the error was encountered, the number of errors that were found for the same error code in the file, and a probable solution to resolve the error.

The Detailed Report lists the detailed description of the error with reference to the object, class, association details included in the objects.csv, links.csv, and model.xml files.

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